

# FINAL EXPLANATION: MATHEMATICS GRADE 10

## *Student Assessment Document*

### FINAL EXPLANATION: MATHEMATICS GRADE 10: 4.2 PROBABILITY I

#### Student Assessment Document

|              |       |
|--------------|-------|
| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

#### INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Use what you learned about Mama Njeri's mandazi stall at Gikomba Market to answer the driving question: How can we use mathematics to measure and predict the likelihood of uncertain everyday events, so we can make smarter decisions even when we cannot be certain of the outcome? Write your answers in the space under each prompt. Show your working for every calculation and write probabilities as fractions or decimals between 0 and 1. Use words from the unit, such as random experiment, sample space, event, relative frequency, mutually exclusive and independent. Always link your answer back to Mama Njeri's stall.

#### Part 1: Why does the fraction of good days settle down?

Mama Njeri kept a 60-day notebook. A 'good day' is a day when she sells at least 80% of her stock. Suppose 24 of the 60 days were good days. (a) Calculate the experimental probability of a good day. (b) In your own words, explain what this number tells Mama Njeri and what it does NOT tell her about tomorrow. (c) Your class plotted running relative frequencies for coins and bottle caps. Use that evidence to explain why the fraction of good days in her notebook becomes steadier the longer she keeps records, even though each single day is unpredictable.

*Write your answer here:*

## Part 2: Reasoning before the stall opens

**(a) List the sample space when two fair coins are tossed together. (b) Use your sample space to find the theoretical probability of getting exactly one head. (c) Explain how theoretical probability helps us understand why experimental relative frequencies settle down over many trials, and why Mama Njeri can reason about probability even before she opens her stall.**

*Write your answer here:*

### Part 3: Events that overlap

Over 20 days in Nakuru, a seller recorded days with HIGH SALES (H) and days with HEAVY RAIN (R). There were 8 days with high sales and 7 days with heavy rain. On 3 of the days there was both high sales and heavy rain. (a) How many of the 20 days had high sales OR heavy rain (or both)? Use a Venn diagram to help you. (b) Write your answer to (a) as a probability  $P(H \cup R)$ . (c) A student finds the number of days by working out  $8 + 7$ . Is this correct? Explain, and write the Addition Law of Probability for any two events A and B. (d) Explain how the Addition Law changes when two events are mutually exclusive. Give one example from Mama Njeri's stall.

*Write your answer here:*

### Part 4: One day after another

Suppose the probability that any day is a good day for Mama Njeri is 0.4, and each day is independent of the others. (a) Draw a two-day probability tree diagram. Label every branch and every path with its probability. (b) Find the probability of two good days in a row, the probability of two bad days in a row, and the probability of exactly one good day in the two days. (c) Explain how you could check that your path probabilities are all correct. (d) A classmate says that independent events and mutually exclusive events mean the same thing. Do you agree? Explain the difference using an example from Mama Njeri's stall.

*Write your answer here:*

### **Part 5: Advising Mama Njeri and answering the driving question**

**Write a short recommendation to Mama Njeri about how to plan her daily batch of mandazi. (a) Use at least three tools from this unit (for example experimental probability, the Addition Law, the Multiplication Law or a tree diagram), and use numbers from your earlier answers. (b) Explain why your advice is a better strategy than guessing, even though probability cannot tell her what will happen tomorrow. (c) Using your recommendation, answer the driving question in two or three sentences.**

Write your answer here:

| RUBRIC                                                       |                                                                                                                                                                            |                                                                                                                                                   |                                                                                                                                    |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Criterion                                                    | Excellent (4)                                                                                                                                                              | Proficient (3)                                                                                                                                    | Developing (1–2)                                                                                                                   |
| Calculating and interpreting probabilities                   | All probabilities are calculated correctly from the data or sample space, shown with working, and interpreted as long-run likelihoods that do not guarantee a single day.  | Calculations are mostly correct with some working; interpretation is mostly correct but may imply certainty for one day.                          | Calculations contain errors or lack working; interpretation of probability is unclear or treats it as a prediction of one outcome. |
| Explaining stabilisation of relative frequency               | Clearly explains, using class coin and bottle-cap evidence, why more trials make relative frequency settle near the true probability while single days stay unpredictable. | States that more trials give a steadier fraction and refers to evidence, but the reason is only partly explained.                                 | Does not explain why the fraction settles, or does not use evidence from the experiments.                                          |
| Using the Addition and Multiplication Laws and tree diagrams | Chooses and applies the correct law, explains double counting, builds a fully labelled tree diagram, and clearly distinguishes independent from mutually exclusive events. | Applies the laws and tree diagram mostly correctly; the explanation of double counting or of independent versus mutually exclusive is incomplete. | Uses the wrong law, has an incomplete or unlabelled tree diagram, or confuses independent and mutually exclusive events.           |
| Using probability to make a decision                         | Gives a justified recommendation to Mama Njeri that uses several tools and specific numbers, and answers the driving question including the role of                        | Gives a reasonable recommendation using some tools and numbers, and a partly complete answer to the driving question.                             | Recommendation is vague or not supported by probability evidence, and the driving question is not answered.                        |

|                            |                                                                           |                                                             |                                                              |
|----------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------|
|                            | uncertainty.                                                              |                                                             |                                                              |
| Mathematical communication | Uses unit vocabulary accurately and writes in clear, organised sentences. | Uses most vocabulary correctly; writing is generally clear. | Vocabulary is missing or misused; writing is hard to follow. |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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